

1. As a Product

It delivers computing potential across networks of Hardware.

It enables the Hardware to deliver the expected functionality.

Example:

- MS Word is a software product used by people to create documents.
- Mobile banking app helps customers manage their accounts.

It acts as an information transformer because it produces, manages, acquires, modifies, displays, or transmits information.

2. As a Vehicle for Delivering a Product

It provides system functionality (e.g., payroll system).

It controls other software (e.g., an operating system).

It helps build other software (e.g., software tools).

Example:

- The firmware in a washing machine controls the hardware and delivers the washing functionality.
- Air traffic control software is not the end product itself but the medium through which the safety service is delivered.

Objectives of Software Engineering

Maintainability: It should be feasible for the software to evolve to meet changing requirements.

Efficiency: The software should not make wasteful use of computing devices such as memory, processor cycles, etc.

Correctness: A software product is correct if the different requirements specified in the SRS Document have been correctly implemented.